glu Presentations

PDF slides with Markdown and Lua

Patrick — March 2026

boxes & glue

Why glu for slides?

- **Markdown** — easy to write
- **Lua** — dynamic and programmable
- **CSS** — full control over layout
- **PDF** — universal output format

> *Less overhead than LaTeX/Beamer, more control than Marp.*

How it works

A Markdown file with YAML frontmatter:

```
---
title: My Talk
css: slides.css
---
```

Each `## heading` starts a new slide (via CSS `break-before: always`).

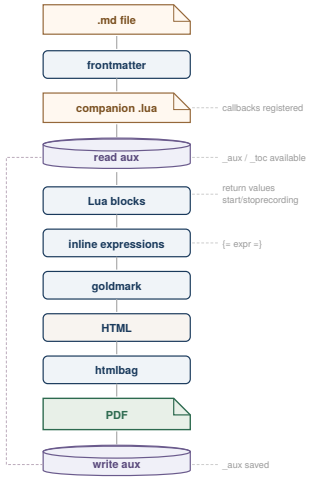
Lua code embedded directly: **4** is four.

Lua blocks

This talk covers **4** topics:

1. Text formatting
2. Tables and lists
3. SVG graphics
4. Dynamic content

The Markdown pipeline



Tables

Feature	Supported
Markdown syntax	Yes
CSS styling	Yes
Lua logic	Yes
SVG graphics	Yes
Page numbers	Yes

Callbacks for recurring elements

In the companion file `slides.lua`:

- **page_init** — page number, accent line, logo
- **pre_shipout** — final adjustments before rendering
- **document_end** — post-processing

```
frontend.add_callback("page_init", "slide_number",
    function(doc, page, pagenum, pageinfo)
        -- page number in the bottom right
    end)
```

Lua blocks: syntax

This is what a Lua block looks like in Markdown:

````
```{lua}
-- executed and the result is inserted
items = { "Item 1", "Item 2" }
return "There are " .. #items .. " items."
```
````

Inline expressions with **{= … =}** are replaced directly.

To only **display** code (without executing): **{!lua}** instead of {lua}.

Verbatim blocks

Want to display code blocks with ``` verbatim? Simply wrap them with more backticks:

````
```{lua}
return "Hello from Lua!"
```
````

The outer 5 backticks protect the inner block — it is **displayed**, not executed.

Lua blocks: live example

3 kinds of fruit: Apples, Pears, Cherries

And the same as a non-executed block with {!lua}:

```
example = { "Apples", "Pears", "Cherries" }
return #example .. " kinds of fruit"
```

Thank you!

Source code: `playground/slides.md`

Back to: Why glu for slides?

Run with:

```
glu slides.md
```